

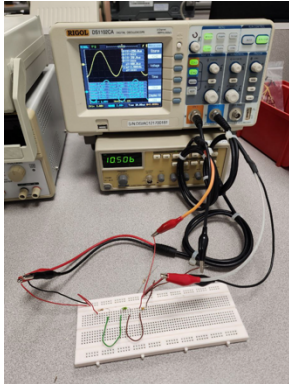
EEP2 – Week 5

14/2/2025 – E4A-06-03

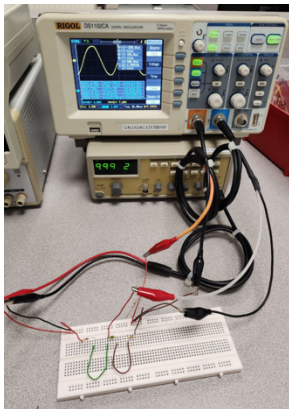
A0287717M – Brians Tjipto Meidianto

Lab

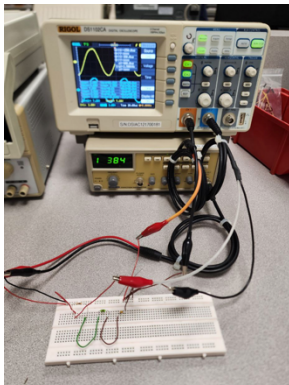
Activity #1



$$V_C = 6.63V$$

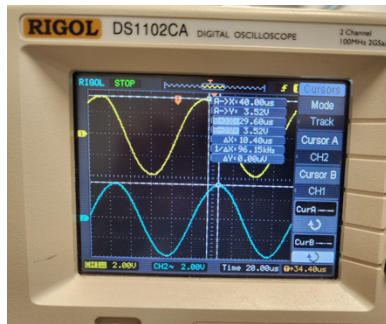


$$V_L = 4.74V$$



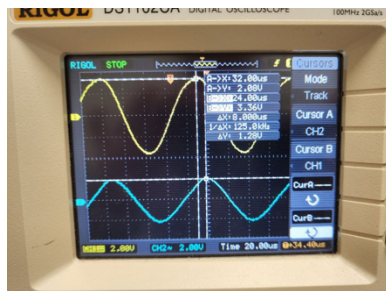
$$V_R = 7.13V$$

Capacitor:



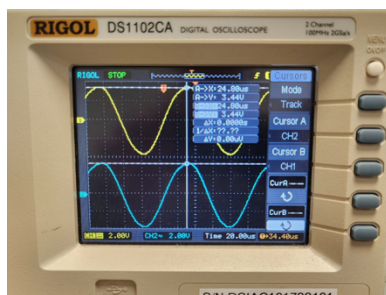
$$10.4\mu s$$

Inductor:



$$8\mu s$$

Resistor:



$$0\mu s$$

Calculate Angular Frequency

$$\omega = 2\pi f = 2\pi(10 \times 10^3) = 62831.85 \text{ rad/s}$$

Convert Phase Shifts to Phase Angles

$$\theta = \omega \times \text{time shift}$$

$$\text{Resistor: } \theta_R = 0 \text{ deg}$$

Capacitor: $\theta_C = 62831.85 \times 10.4 \times 10^{-6} = 0.653 \text{ rad} = 37.4 \text{ deg}$

Inductor: $\theta_L = 62831.85 \times 8 \times 10^{-6} = 0.503 \text{ rad} = 28.8 \text{ deg}$

Convert to Phasor Form

$$\hat{V} = V_{rms} e^{j\theta}$$

$$\hat{V}_R = 7.13 \angle 0$$

$$\hat{V}_L = 4.74 \angle 28.8$$

$$\hat{V}_C = 6.63 \angle 37.4$$

$$\hat{V}_S = 10 \angle 0$$

Verify KVL Using Phasors

$$\hat{V}_R = 7.13 + j0$$

$$\hat{V}_L = 4.74 \cos 28.8 + j4.74 \sin 28.8 = 4.15 + j2.29$$

$$\hat{V}_C = 6.63 \cos -37.4 + j6.63 \sin -37.4 = 5.27 - j4.03$$

Sum

$$16.55 - j1.74$$

Magnitude

$$\sqrt{16.55^2 + (-1.74)^2} \approx 16.64V$$

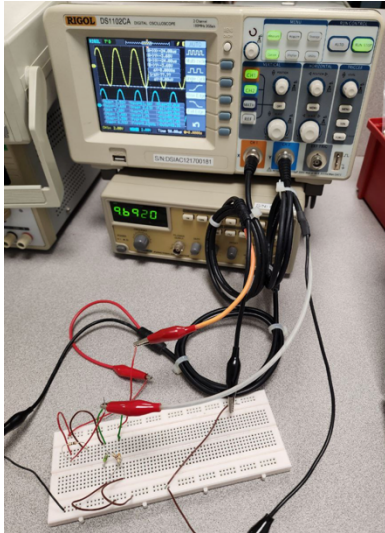
Phase angle:

$$\theta_{total} = \tan^{-1} \left(\frac{-1.74}{16.55} \right) = -6 \text{ deg}$$

Since magnitude is approx. 16V and phase angle is close to 0deg means it satisfies KVL

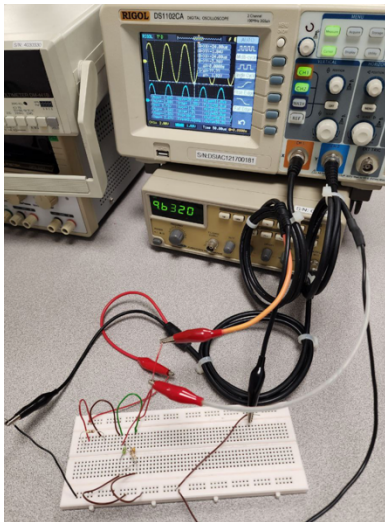
Activity #2

Resistor:



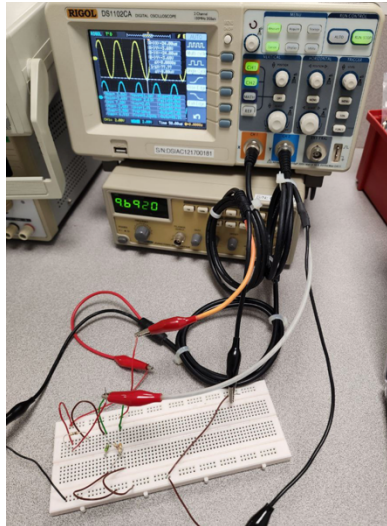
3.45A

Inductor:



3.59A

Capacitor:



3.58A

Source:

7.04A

$$3.45 + ((3.58+3.59)/2) = 7.035A$$

7.035A is approx. 7.04, hence satisfies KCL